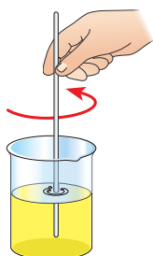


Chapter 1 – Introduction to Science

Basic practical skills

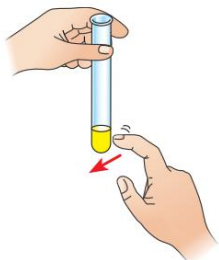
Mixing solutions



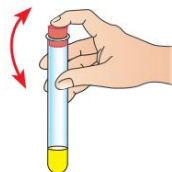
Method 1:
Use a glass rod to stir
the mixed solution



Method 2:
Gently shake the
test tube

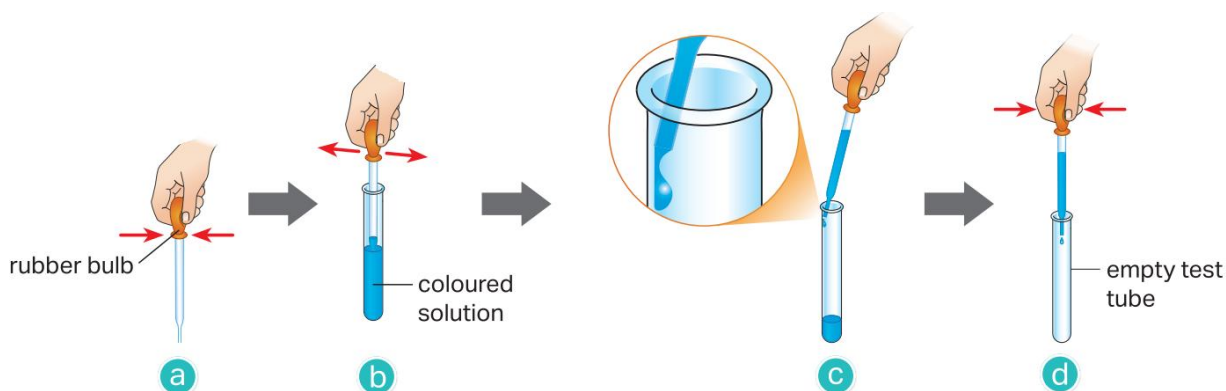


Method 3:
Tap the bottom of the test
tube with the index finger



Method 4:
Stopper the test tube
and shake well

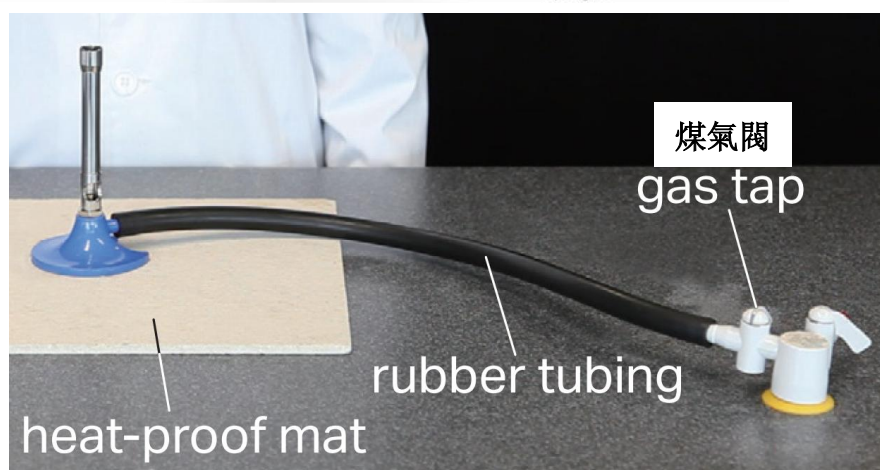
Transferring small amount of solution



- Squeeze 擠 the rubber bulb 橡膠球 of dropper and release to draw solution
- Then remove excess solution by touching the wall of test tube
- Release the bulb to release solution.

Heating with Bunsen burner 使用本生燈加熱

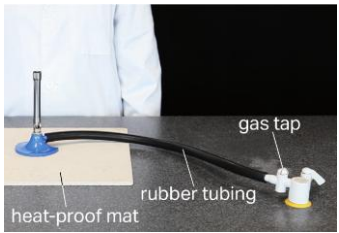
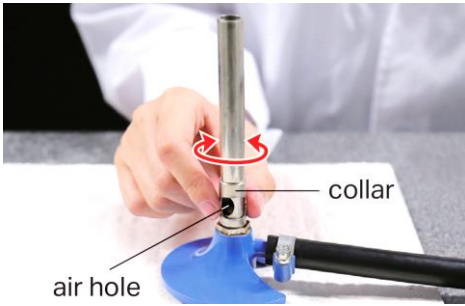
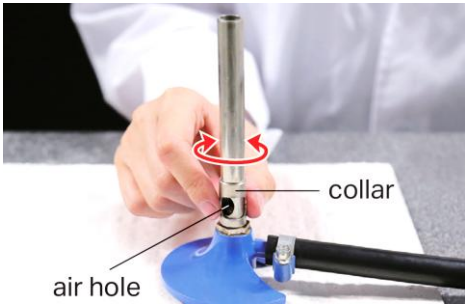
Structure of Bunsen burner



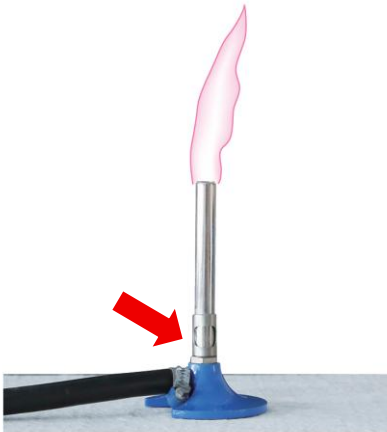
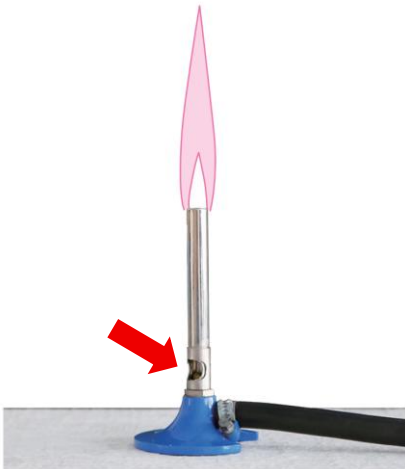
Setup of Bunsen burner

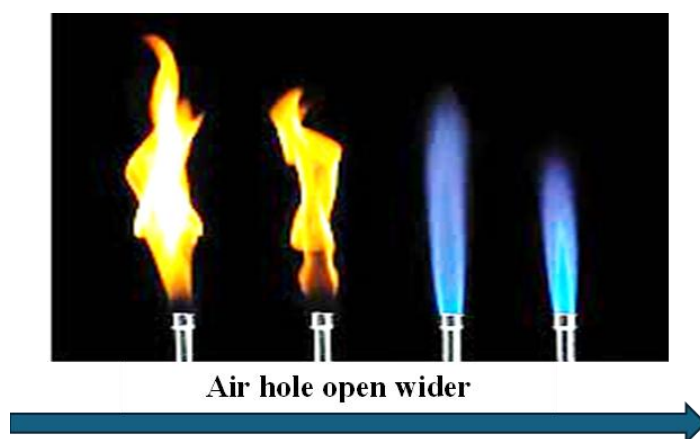


- Light a Bunsen burner

Step	Action	Key Details	
1	Setup	Place Bunsen burner on a heat-proof mat: to <u>protect the bench from heat</u>	
2	Adjust collar	Turn collar to close the air hole: <u>reduce the amount of air that mixes with the fuel gas</u> → create a bright yellow flame that is stable and easier to see	
3	Ignition 點燃	a) Position gas lighter above chimney	
		b) Press lighter button c) Slowly open gas tap	
		d) Adjust the flame by slowly opening the air hole. Do not open the air hole fully!	

- Openness of air hole and appearance of the flame

State of air hole	Air Hole Closed	Air Hole Open
Amount of air mix with fuel	Less	More
State of burning	Gentle 溫和	Intense 強烈
Temperature	Lower	Higher
Appearance of the flame	Luminous Flame 發光焰 	Non-Luminous Flame 無光焰 
Colour	Yellow	Blue
Shape	Irregular	Regular
Noise Level	Quiet	Noisy



Strike back 回擊



- If there are problems in operating the Bunsen burner, e.g. opening the air hole fully or any debris stuck inside chimney, the **Bunsen flame may burn inside the chimney**, this is called **strike back** and you may observe a **green flame** and hear **hissing sounds**

**You SHOULD NOT touch the chimney or collar and
Should turn the gas tap off immediately!**

A. Multiple Choice Questions (MCQ)

1. What is the first step when using a Bunsen burner?
 - A. Turn on the gas tap.
 - B. Place the Bunsen burner on a heat-proof mat.
 - C. Open the air hole fully.
 - D. Light the flame.
2. What happens if you open the air hole fully?
 - A. The flame becomes yellow and quiet.
 - B. The flame becomes blue and noisy.
 - C. The flame burns inside the chimney, causing strike back.
 - D. The flame extinguishes immediately.

3. What is the purpose of closing the air hole before lighting the Bunsen burner?
 - A. To make the flame blue.
 - B. To ensure a luminous flame for easier ignition.
 - C. To increase the temperature of the flame.
 - D. To reduce noise levels.
4. Which of the following conditions indicates a non-luminous flame?
 - A. Yellow color, irregular shape, quiet.
 - B. Blue color, regular shape, noisy.
 - C. Green color, hissing sound.
 - D. Yellow color, regular shape, quiet.
5. What should you do if you observe a green flame and hear hissing sounds?
 - A. Adjust the collar to close the air hole.
 - B. Turn off the gas tap immediately.
 - C. Touch the chimney to fix the problem.
 - D. Fully open the air hole.

B. Fill-in-the-Blank Questions

1. Place the Bunsen burner on a _____ mat.
2. Before lighting the Bunsen burner, turn the collar to _____ the air hole.
3. A luminous flame is _____ in color and has an _____ shape.
4. A non-luminous flame is _____ in color, has a _____ shape, and makes a _____ sound.
5. To turn off the Bunsen burner, close the _____ and the _____.
6. If _____ occurs, you may observe a green flame and hear hissing sounds.

C. True or False

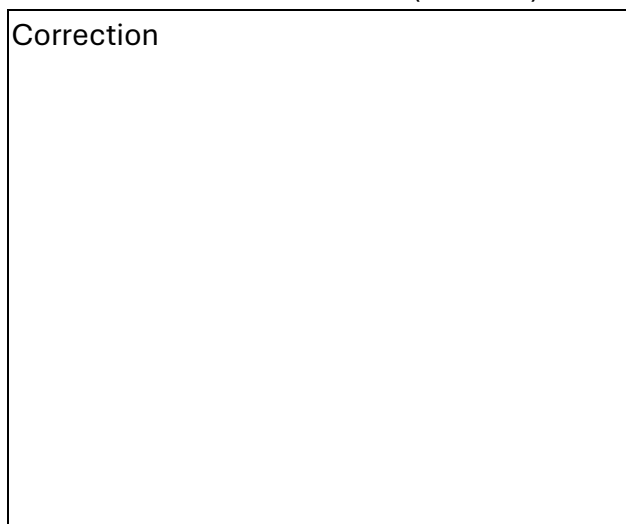
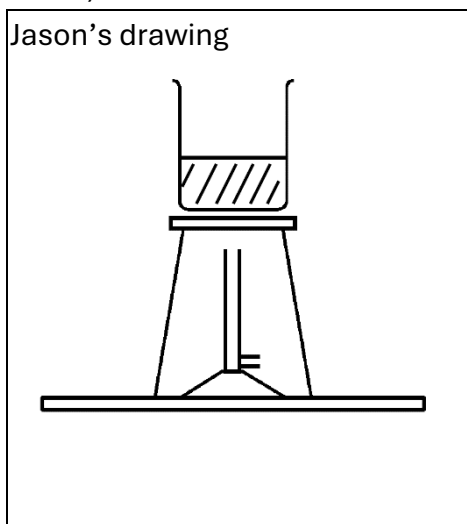
- | | |
|---|-----|
| 1. You should always fully open the air hole when lighting the Bunsen burner. | T/F |
| 2. A blue flame is hotter than a yellow flame. | T/F |
| 3. If strike back occurs, you should touch the chimney to fix it. | T/F |
| 4. A luminous flame is suitable for heating objects. | T/F |
| 5. The air hole should be closed when turning off the Bunsen burner. | T/F |

D. Long Question

Jason used the following set-up to boil some water in the laboratory.



- (a) Jason drew the set-up as shown below. There are TWO mistakes in his drawing. Circle them and draw the correct diagram in the space provided. (Note: Labels can be omitted.) (4 marks)



- (b) When we boil the water, should we open or close the air hole why? (3 marks)

- (c) What is the function of heat proof mat? (1 marks)

Answer Key

A. Multiple Choice Questions (MCQ)

1. B
2. C
3. B
4. B
5. B

B. Fill-in-the-Blank Questions

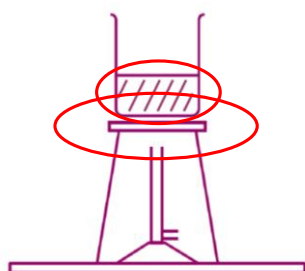
1. heat-proof
2. close
3. yellow , irregular
4. blue , regular , noisy
5. air hole , gas tap
6. strike back

C. True or False

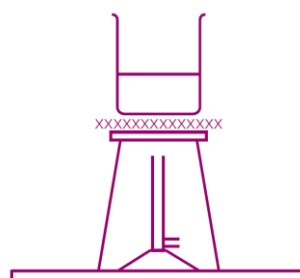
1. False
2. True
3. False
4. False
5. True

D. Long Question

(a) Jason's drawing



Correction



- | | |
|---|-----|
| (b) We should open the air hole | (1) |
| because it will produce non-luminous flame | (1) |
| which has higher temperature and heat up the water faster | (1) |
| (c) Protect the bench from heat | (1) |